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SHORT-PAPER

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Published: 21 October 2023

[Citation in BibTeX format](#)

CIKM '23: The 32nd ACM International Conference on Information and Knowledge Management
October 21 - 25, 2023
Birmingham, United Kingdom

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Spatio-Temporal Adaptive Embedding Makes Vanilla Transformer SOTA for Traffic Forecasting

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ABSTRACT

With the rapid development of the Intelligent Transportation System (ITS), accurate traffic forecasting has emerged as a critical challenge. The key bottleneck lies in capturing the intricate spatio-temporal traffic patterns. In recent years, numerous neural networks with complicated architectures have been proposed to address this issue. However, the advancements in network architectures have encountered diminishing performance gains. In this study, we present a novel component called *spatio-temporal adaptive embedding* that can yield outstanding results with vanilla transformers. Our proposed Spatio-Temporal Adaptive Embedding transformer (STAEformer) achieves state-of-the-art performance on five real-world traffic forecasting datasets. Further experiments demonstrate that spatio-temporal adaptive embedding plays a crucial role in traffic forecasting by effectively capturing intrinsic spatio-temporal relations and chronological information in traffic time series.

CCS CONCEPTS

• Information systems → Spatial-temporal systems; • Computing methodologies → Artificial intelligence.

KEYWORDS

traffic forecasting, spatio-temporal embedding, transformer

ACM Reference Format:

Hangchen Liu, Zheng Dong, Renhe Jiang, Jiewen Deng, Jinliang Deng, Quanjun Chen, and Xuan Song. 2023. Spatio-Temporal Adaptive Embedding Makes Vanilla Transformer SOTA for Traffic Forecasting. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge*

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CIKM '23, October 21–25, 2023, Birmingham, United Kingdom

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ACM ISBN 979-8-4007-0124-5/23/10...\$15.00

<https://doi.org/10.1145/3583780.3615160>

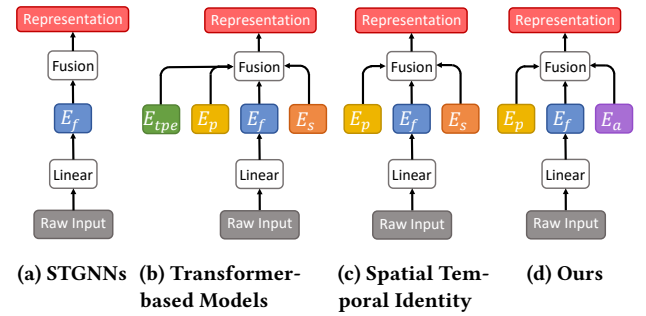


Figure 1: Input Embeddings in Different Traffic Models.

Management (CIKM '23), October 21–25, 2023, Birmingham, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3583780.3615160>

1 INTRODUCTION

Traffic forecasting [16] aims to predict the future traffic time series in road networks based on historical observations. In recent years, the success of deep learning models has been notable, primarily attributable to their ability to capture the inherent spatio-temporal dependencies in the traffic system. Among them, Spatio-Temporal Graph Neural Networks (STGNNs) [19, 38] and Transformer-based models [12, 14, 36, 42] become very popular for their outstanding performance. Researchers have devoted considerable effort to developing fancy and complicated models for traffic forecasting, such as novel graph convolutions [2, 3, 9–11, 13, 18, 21, 26, 29, 31, 37, 41], learning graph structures [15, 23, 32, 34, 35, 40], efficient attention mechanisms [5, 20, 33, 43, 44], and other methods [4, 8, 17, 22, 25, 30]. Nonetheless, the advancements in network architectures have encountered diminishing performance gains, prompting a shift in focus from complicated model designs towards effective representation techniques for the data itself.

In light of this, in this study, we focus on *input embedding*, a widely-used, simple, yet powerful representation technique, that is often overlooked by many researchers in terms of its effectiveness. Specifically, it adds an embedding layer on the input, providing multiple types of embeddings for the model backbone. Figure 1 presents a comparative analysis of the utilized embeddings by previous models. STGNNs mainly use feature embedding E_f , i.e. a transformation to project the raw input into hidden space. Transformer-based models require additional knowledge such as temporal positional

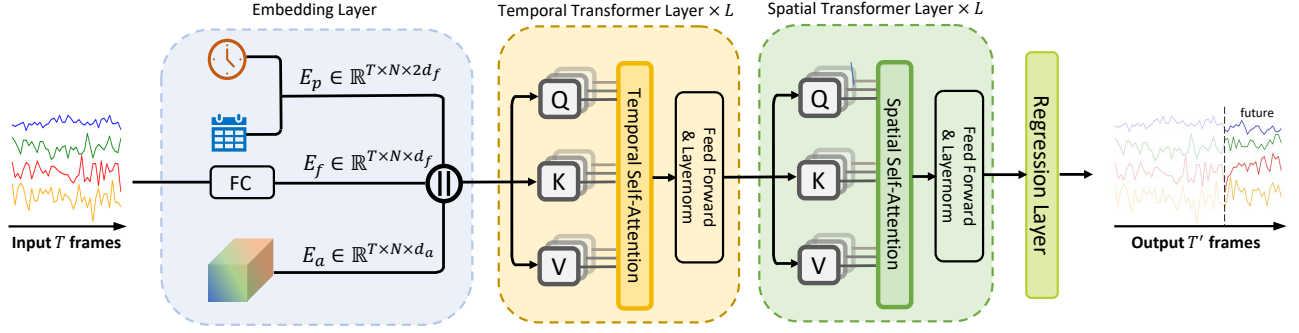


Figure 2: The Architecture of Spatio-Temporal Addaptive Embeding transformer (STAEformer).

encoding E_{tpe} and periodicity (daily, weekly, monthly) embedding E_p due to the attention mechanism's inability to preserve the positional information of the time series. Recent models, including PDFormer [14], GMAN [42] and STID [24], apply spatial embedding E_s . Notably, STID [24] is among the few studies that explore these embeddings. It employs spatial embedding and temporal periodicity embedding with a simple Multi-layer Perceptron (MLP) and achieves remarkable performance.

To further enhance the representation effectiveness, we propose a novel *spatio-temporal adaptive embedding* E_a and apply it on the vanilla Transformer [28] along with E_p and E_f , which is shown in Figure 1d. In detail, the raw input goes through an embedding layer to obtain the input embedding, which is fed into temporal and spatial transformer layers followed by a regression layer to make predictions. Our proposed model, named Spatio-Temporal Addaptive Embeding transformer (STAEformer), has a far more concise architecture but achieves the state-of-the-art (SOTA) performance. In our model, E_a plays a crucial role by effectively capturing intrinsic spatio-temporal relations and chronological information in traffic time series. Experiments and analyses on five real-world traffic datasets prove that our proposed E_a can make vanilla transformer SOTA for traffic forecasting.

2 PROBLEM DEFINITION

Given the traffic series $X_{t-T+1:t}$ in the previous T time frames, traffic forecasting aims to infer the traffic data in the future T' frames formulated as $[X_{t-T+1}, \dots, X_t] \rightarrow [X_{t+1}, \dots, X_{t+T'}]$, where each frame $X_i \in \mathbb{R}^{N \times d}$, N is the number of spatial nodes, and $d=1$ is the input dimension for the traffic volume.

3 METHODOLOGY

As shown in Figure 2, our model consists of an embedding layer, vanilla transformers applied along temporal axis as the temporal transformer layer and along spatial axis as the spatial transformer layer, then a regression layer.

3.1 Embedding Layer

To keep the native information in the raw data, we utilize a fully connected layer to obtain the feature embedding $E_f \in \mathbb{R}^{T \times N \times d_f}$:

$$E_f = FC(X_{t-T+1:t}) \quad (1)$$

where d_f is the dimension of the feature embedding, and $FC(\cdot)$ indicates a fully connected layer.

We denote the learnable day-of-week embedding dict as $T_w \in \mathbb{R}^{N_w \times d_f}$, and the timestamps-of-day embedding dict as $T_d \in \mathbb{R}^{N_d \times d_f}$, where $N_w = 7$ denotes the number of days in a week. In our case, $N_d = 288$ is the number of timestamps in each day. Denoting $W^t \in \mathbb{R}^T$, $D^t \in \mathbb{R}^T$ as the day-of-week data and the timestamp-of-day data for the traffic time series in $t-T+1:t$, we use them as indices to extract the corresponding day-of-week embedding $E_w^t \in \mathbb{R}^{T \times d_f}$ and timestamp-of-day embedding $E_d^t \in \mathbb{R}^{T \times d_f}$ from the embedding dicts. By concatenating and broadcasting them, we get the periodicity embedding $E_p \in \mathbb{R}^{T \times N \times 2d_f}$ for the traffic time series.

On one hand, it's intuitive that the temporal relation is not only decided by periodicity but also affected by the chronological order in the traffic time series. For example, a time frame in traffic time series should be more similar to the time frames nearby. On the other hand, the time series from different sensors tend to have different temporal patterns. Thus, instead of adopting a pre-defined or dynamic adjacency matrix for spatial relation modeling, we designed a spatio-temporal adaptive embedding $E_a \in \mathbb{R}^{T \times N \times d_a}$ to capture intricate spatio-temporal relation in a uniform way. In particular, E_a is shared across different traffic time series.

By concatenating the embeddings above, we obtain the hidden spatio-temporal representation $Z \in \mathbb{R}^{T \times N \times d_h}$ as follows:

$$Z = E_f || E_p || E_a \quad (2)$$

where the hidden dimension d_h equals $3d_f + d_a$.

3.2 Transformer and Regression Layer

We apply vanilla transformers along temporal and spatial axes to capture intricate traffic relations. Given hidden spatio-temporal representation $Z \in \mathbb{R}^{T \times N \times d_h}$ with T frames and N spatial nodes, we obtain the query, key and value matrices through temporal transformer layers as:

$$Q^{(te)} = ZW_Q^{(te)}, K^{(te)} = ZW_K^{(te)}, V^{(te)} = ZW_V^{(te)} \quad (3)$$

where $W_Q^{(te)}, W_K^{(te)}, W_V^{(te)} \in \mathbb{R}^{d_h \times d_h}$ are learnable parameters. Then we calculate the self-attention score as:

$$A^{(te)} = \text{Softmax} \left(\frac{Q^{(te)} K^{(te)\top}}{\sqrt{d_h}} \right) \quad (4)$$

Table 1: Performance on METR-LA and PEMS-BAY.

Datasets	Metric	HI	GWNet	DCRNN	AGCRN	STGCN	GTS	MTGNN	STNorm	GMAN	PDFormer	STID	STAEformer
METR-LA	Horizon 3 (15 min)	MAE	6.80	2.69	2.67	2.85	2.75	2.69	2.81	2.80	2.83	2.82	2.65
		RMSE	14.21	5.15	5.16	5.53	5.29	5.16	5.57	5.55	5.45	5.53	5.11
		MAPE	16.72%	6.99%	6.86%	7.63%	7.10%	7.12%	6.89%	7.40%	7.41%	7.75%	6.85%
	Horizon 6 (30 min)	MAE	6.80	3.08	3.12	3.20	3.15	3.14	3.05	3.18	3.12	3.20	3.19
		RMSE	14.21	6.20	6.27	6.52	6.35	6.13	6.59	6.49	6.46	6.57	6.00
		MAPE	16.72%	8.47%	8.42%	9.00%	8.62%	8.62%	8.16%	8.47%	8.73%	9.19%	8.13%
	Horizon 12 (60 min)	MAE	6.80	3.51	3.54	3.59	3.60	3.59	3.47	3.44	3.62	3.55	3.34
		RMSE	14.20	7.28	7.47	7.45	7.43	7.44	7.21	7.51	7.35	7.47	7.02
		MAPE	10.15%	9.96%	10.32%	10.47%	10.35%	10.25%	9.70%	10.24%	10.07%	10.91%	9.70%
PEMS-BAY	Horizon 3 (15 min)	MAE	3.06	1.30	1.31	1.35	1.36	1.37	1.33	1.35	1.32	1.31	1.31
		RMSE	7.05	2.73	2.76	2.88	2.88	2.92	2.80	2.82	2.90	2.79	2.78
		MAPE	6.85%	2.71%	2.73%	2.91%	2.86%	2.85%	2.81%	2.76%	2.87%	2.78%	2.76%
	Horizon 6 (30 min)	MAE	3.06	1.63	1.65	1.67	1.70	1.72	1.66	1.65	1.64	1.64	1.62
		RMSE	7.04	3.73	3.75	3.82	3.84	3.86	3.77	3.77	3.82	3.79	3.68
		MAPE	6.84%	3.73%	3.71%	3.81%	3.79%	3.88%	3.75%	3.66%	3.74%	3.71%	3.62%
	Horizon 12 (60 min)	MAE	3.05	1.99	1.97	1.94	2.02	2.06	1.95	1.92	1.91	1.91	1.88
		RMSE	7.03	4.60	4.60	4.50	4.63	4.60	4.50	4.45	4.43	4.42	4.34
		MAPE	6.83%	4.71%	4.68%	4.55%	4.72%	4.88%	4.62%	4.46%	4.52%	4.55%	4.41%

where $A^{(te)} \in \mathbb{R}^{N \times T \times T}$ captures the temporal relations in different spatial nodes. Finally, we obtain the output of temporal transformer $Z^{(te)} \in \mathbb{R}^{T \times N \times d_h}$ as:

$$Z^{(te)} = A^{(te)} V^{(te)}. \quad (5)$$

Similarly, the spatial transformer layer performs as:

$$Z^{(sp)} = \text{SelfAttention}(Z^{(te)}) \quad (6)$$

where *SelfAttention* follows Equation 3, 4, 5, and $Z^{(sp)} \in \mathbb{R}^{T \times N \times d_h}$ is the output of the spatial transformer. Notably, we also apply layer normalization, residual connection and multi-head mechanism.

Finally, we leverage the output of the spatio-temporal transformer layers $Z' \in \mathbb{R}^{T \times N \times d_h}$ to generate predictions. The regression layer can be formulated as:

$$\hat{Y} = FC(Z') \quad (7)$$

where $\hat{Y} \in \mathbb{R}^{T' \times N \times d}$ is the prediction, T' is the horizon of prediction. d is the dimension of the output features which equals 1 in our case. Thus, the fully-connected layer regresses the dimension from $T \times d_h$ in Z' to $T' \times (d = 1)$ in $\hat{Y}$.

4 EXPERIMENTS

4.1 Experimental Setup

Settings. Our method is verified on five traffic forecasting benchmarks (METR-LA [19], PEMS-BAY [19], PEMS04 [26], PEMS07 [26], and PEMS08 [26]). The traffic data in the five datasets are aggregated to 5 minutes. Thus there are 12 frames in each hour. More details are shown in Table 2. METR-LA and PEMS-BAY are divided into the training, validation, and test sets in a fraction of 7:1:2. PEMS04, PEMS07 and PEMS08 are divided in a fraction of 6:2:2. In fact, the

Table 2: Summary of Datasets.

Dataset	#Sensors (N)	#Timesteps	Time Range
METR-LA	207	34,272	03/2012 - 06/2012
PEMS-BAY	325	52,116	01/2017 - 05/2017
PEMS04	307	16,992	01/2018 - 02/2018
PEMS07	883	28,224	05/2017 - 08/2017
PEMS08	170	17,856	07/2016 - 08/2016

Table 3: Performance on PEMS04, PEMS07, and PEMS08.

Dataset	PEMS04			PEMS07			PEMS08		
Metric	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
HI	42.35	61.66	29.92%	49.03	71.18	22.75%	36.66	50.45	21.63%
GWNet	18.53	29.92	12.89%	20.47	33.47	8.61%	14.40	23.39	9.21%
DCRNN	19.63	31.26	13.59%	21.16	34.14	9.02%	15.22	24.17	10.21%
AGCRN	19.38	31.25	13.40%	20.57	34.40	8.74%	15.32	24.41	10.03%
STGCN	19.57	31.38	13.44%	21.74	35.27	9.24%	16.08	25.39	10.60%
GTS	20.96	32.95	14.66%	22.15	35.10	9.38%	16.49	26.08	10.54%
MTGNN	19.17	31.70	13.37%	20.89	34.06	9.00%	15.18	24.24	10.20%
STNorm	18.96	30.98	12.69%	20.50	34.66	8.75%	15.41	24.77	9.76%
GMAN	19.14	31.60	13.19%	20.97	34.10	9.05%	15.31	24.92	10.13%
PDFormer	18.36	30.03	12.00%	19.97	32.95	8.55%	13.58	23.41	9.05%
STID	18.38	29.95	12.04%	19.61	32.79	8.30%	14.21	23.28	9.27%
STAEformer	18.22	30.18	11.98%	19.14	32.60	8.01%	13.46	23.25	8.88%

Table 4: Ablation Study on PEMS04, PEMS07 and PEMS08.

Dataset	PEMS04			PEMS07			PEMS08		
Metric	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
w/o E_a	21.63	34.88	14.26%	22.37	37.21	9.25%	15.00	25.85	9.74%
w/o E_p	18.92	30.74	12.27%	20.21	33.59	8.40%	15.00	24.05	9.56%
w/o T -Trans	18.50	30.34	12.24%	19.87	33.67	8.29%	13.83	23.79	9.09%
w/o ST -Trans	25.17	39.89	18.00%	27.71	42.99	17.61%	19.96	31.47	18.53%
STAEformer	18.22	30.29	12.01%	19.14	32.60	8.01%	13.46	23.25	8.88%

performance of our model is not sensitive to the hyper-parameters. For more details, the embedding dimension d_f is 24 and the d_a is 80. The number of layers L is 3 for both spatial and temporal transformers. The number of heads is 4. We set the input and prediction length to be 1 hour, namely, $T=T'=12$. Adam is chosen as the optimizer with the learning rate decaying from 0.001, and the batch size is 16. We apply an early-stop mechanism if the validation error converges within 30 continuous steps. We use three widely used metrics for traffic forecasting task, i.e, MAE, RMSE and MAPE. Following previous work, we select the average performance of all predicted 12 horizons on the PEMS04, PEMS07 and PEMS08 datasets. To evaluate the METR-LA and PEMS-BAY datasets, we compare

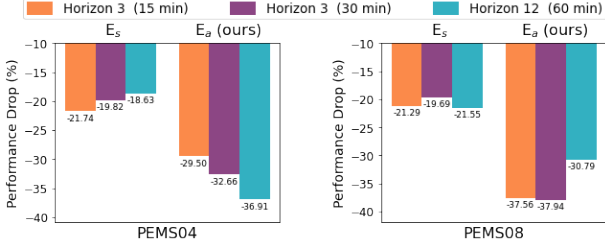


Figure 3: By shuffling the input along the temporal axis, our spatio-temporal adaptive embedding shows a more dramatic performance decrease than spatial embedding, indicating its capability in capturing chronological information.

the performance on horizon 3, 6 and 12 (15, 30, and 60 min). The code is available at <https://github.com/XDZhelheim/STAEformer>.

Baselines. In this study, we compare our proposed method against several widely used baselines in the field. HI [6] is a typical traditional model. We also consider STGNNs such as GWNet [35], DCRNN [19], AGCRN [1], STGCN [38], GTS [23], and MTGNN [34], which employ the embeddings shown in Figure 1(a). Additionally, we examine STNorm [7], which focuses on factorizing traffic time series. While there exist Transformer-based methods for time series forecasting, such as Informer [43], Pyraformer [20], FEDformer [44], and Autoformer [33], they are not specially tailored for short-term traffic forecasting. Hence, we select GMAN [42] and PDFormer [14], which are transformer models targeting the same task as ours. The input embeddings in [42] and [14] follow the configuration in Figure 1(b). Furthermore, we consider STID [24], which enhances the spatio-temporal distinction in traffic time series by utilizing the input embedding depicted in Figure 1(c).

4.2 Performance Evaluation

As shown in Table 1 and 3, our method achieves better performance on most metrics on all five datasets. STAEformer outperforms STGNNs by a large degree without any graph modeling. STNorm and STID also achieve competitive results, while Transformer-based models can better capture the intricate spatio-temporal relations. Compared with PDFormer, the encouraging results indicate STAEformer is a simpler but more effective solution.

4.3 Ablation Study

To evaluate the effectiveness of each part in STAEformer, we conduct ablation studies with 4 variants of our model as follows:

- **w/o E_a .** It removes spatio-temporal adaptive embedding E_a .
- **w/o E_p .** It removes periodicity embedding E_p , including day-of-week and timestamps-of-day embedding.
- **w/o T -Trans.** It removes temporal transformer layers.
- **w/o ST -Trans.** It removes both temporal transformer layers and spatial transformer layers.

Table 4 reveals the significance of various embeddings on the performance of our model. E_p can capture daily and weekly patterns, while the proposed E_a is vital for traffic modeling. Furthermore, the substantial performance degradation observed upon the removal of spatial or temporal transformer layers shows that our proposed embeddings effectively model the inherent spatio-temporal patterns

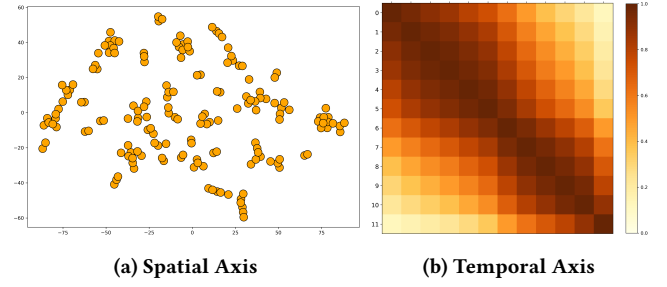


Figure 4: Visualization of Spatio-Temporal Adaptive Embedding E_a on PEMS08.

in the traffic data. Therefore, both spatial and temporal layers are necessary for extracting those features.

4.4 Case Study

Comparison with Spatial Embedding. To validate the effectiveness of spatio-temporal adaptive embedding in capturing chronological order information implied in the T input frames, we conduct more experiments on PEMS04 and PEMS08 datasets. Following [39], we randomly shuffle the raw input along temporal axis T . For comparison, we replace our spatio-temporal adaptive embedding E_a with spatial embedding E_s that was used in [14, 24]. As shown by Figure 3, our model has more severe performance degradation when shuffling the raw input along temporal axis T . It means that spatio-temporal adaptive embedding E_a makes our model more sensitive to the chronological order, while the model with E_s is relatively insensitive. In summary, E_a can better model the chronological information in the raw input and other intricate traffic patterns, which is crucial to the task.

Visualization of Spatio-Temporal Adaptive Embedding. Figure 4 further provides visualizations of our proposed spatio-temporal embedding E_a on the spatial and temporal axes by taking PEMS08 dataset as an example. For the spatial axis, we use t-SNE [27] to get Figure 4a. It shows that the embeddings of different nodes naturally form into clusters, which matches the spatial characteristics of the traffic data. For the temporal axis, we calculate the correlation coefficient across the 12 input frames and draw a heat map as Figure 4b. It shows that each frame is highly correlated to the nearby frames, and the correlation gradually decreases for further frames. This shows that our proposed E_a models the chronological information in the time series correctly.

5 CONCLUSION

In this study, we focus on a basic representation learning technique for traffic time series forecasting, i.e., input embedding. We propose a novel spatio-temporal adaptive embedding that can work on vanilla transformers to achieve the SOTA performance on five traffic benchmarks. Further studies demonstrate that our model can effectively capture intrinsic spatio-temporal dependencies. Instead of designing complicated models, our study shows a promising direction for addressing the challenges in traffic forecasting.

ACKNOWLEDGMENTS

This work was partially supported by the grants of National Key Research and Development Program of China (2021YFB1714400).

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